COMPACTNESS AND INERTIA

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Let E be an elliptic curve over a local field K. The rational points of E with nonsingular reduction form a subgroup $E_0(K)$ of finite index in $E(K)$ [Sil09, Theorem VII.6.1 and Corollary VII.6.2], and E has good reduction if and only if the inertia group acts trivially on its torsion, by the criterion of Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1]. Problem 1.8 proves the finiteness of $E(K)/E_0(K)$ by a compactness argument. Problem 2.12 studies how the inertia group acts on the torsion of a curve with potential good reduction. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A finite extension K of $\mathbb{Q}_p$, with normalized valuation v, ring of integers $\mathcal{O}_K$, uniformizer π and residue field $k = \mathcal{O}_K/\pi\mathcal{O}_K$, a finite field of characteristic p. (Silverman's Chapter VII assumes that K and k are perfect, and here both are.)
- A fixed algebraic closure $\overline{K}$ of K.
- A bar denotes reduction modulo π.
- An elliptic curve E over K, given by a minimal Weierstrass equation $F(X, Y, Z) = 0$, with $F \in \mathcal{O}_K[X, Y, Z]$ homogeneous of degree 3. It embeds E in $\mathbb{P}^2$ as $V_+(F)$.
- The primitive coordinates $[X : Y : Z]$ of $P \in \mathbb{P}^2(K)$: coordinates in $\mathcal{O}_K$, not all in $\pi\mathcal{O}_K$. Every P has them, and they are unique up to a factor in $\mathcal{O}_K^\times$.
- The reduction $\bar{P} \coloneqq [\bar{X} : \bar{Y} : \bar{Z}] \in \mathbb{P}^2(k)$ of $P \in \mathbb{P}^2(K)$, computed from primitive coordinates, hence well defined.
- The reduced curve E_π, defined by $\bar{F} = 0$. If $P \in E(K)$, then $\bar{P} \in E_\pi(k)$.

How to use this. Allow about 25 minutes for each problem. The `Magma` block in Problem 2.12 runs in seconds in the free online calculator. Everything else is by hand.

1. Finiteness by compactness

This section is [Sil09, Exercise 7.6]. The argument uses that k is finite, so it does not apply, for instance, to the completion of the maximal unramified extension of $\mathbb{Q}_p$: a field complete with respect to a discrete valuation, with residue field $\overline{\mathbb{F}}_p$ (item (6)). Yet this is the case that the criterion of Néron–Ogg–Shafarevich needs: Silverman's proof of it uses the finiteness of $E(K^{\mathrm{unr}})/E_0(K^{\mathrm{unr}})$, where K^{unr} is the maximal unramified extension of K, whose residue field is an algebraic closure of k [Sil09, proof of Theorem VII.7.1].

Definition 1.1 (The topology of K)**.** We give K the topology defined by v: the sets $a + \pi^n\mathcal{O}_K$, for $a \in K$ and $n \in \mathbb{Z}$, form a basis, and each of them is both open and closed. With it, K is a Hausdorff topological field.

Definition 1.2 (The subgroup $E_0(K)$)**.** Let E_π^{ns} be E_π minus its singular point, if it has one. Then

$$E_0(K) \coloneqq \{P \in E(K) : \bar{P} \in E_\pi^{\mathrm{ns}}(k)\}.$$

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Definition 1.3 (Regular functions [Poo06, Definition 1.8.3])**.** A function $\psi \in K(E)$ is regular at $Q \in E(\overline{K})$ if $\psi = g/h$ with $g, h \in K[X, Y, Z]$ homogeneous of the same degree and $h(Q) \neq 0$. Then $\psi(Q) \coloneqq g(Q)/h(Q)$, an element of K if $Q \in E(K)$. The functions regular at Q form a discrete valuation ring with fraction field $K(E)$.

Proposition 1.4 ($E_0(K)$ is a subgroup [Sil09, Proposition VII.2.1])**.** *The set $E_0(K)$ is a subgroup of $E(K)$.*

Proposition 1.5 (Compactness [Poo17, §1.1.2])**.** *The ring $\mathcal{O}_K$ is compact: it is complete, and for every n it is covered by the $(\#k)^n$ translates of $\pi^n\mathcal{O}_K$. Equivalently, K is locally compact.*

Proposition 1.6 (Morphisms $E \to E$ over K)**.**
(1) The morphisms $\alpha\colon E \to E$ defined over K are exactly the points $\alpha = [\psi_0 : \psi_1 : \psi_2] \in E(K(E))$, that is, $\psi_0, \psi_1, \psi_2 \in K(E)$, not all zero, with $F(\psi_0, \psi_1, \psi_2) = 0$ [Poo17, Proposition 3.6.5(b)].
(2) If $Q \in E(\overline{K})$ and ψ_0, ψ_1, ψ_2 are regular at Q and not all zero there, then $\alpha(Q) = [\psi_0(Q) : \psi_1(Q) : \psi_2(Q)]$ [Sil09, §I.3].
(3) Replacing each ψ_i by $\phi\psi_i$, for $\phi \in K(E)^\times$, gives the same point of $E(K(E))$, and for every $Q \in E(\overline{K})$ some ϕ makes $\phi\psi_0, \phi\psi_1, \phi\psi_2$ regular at Q and not all zero there: a power of a uniformizer of the discrete valuation ring of Definition 1.3 will do [Sil09, proof of Proposition II.2.1].

Proposition 1.7 (Translations [Sil09, Theorem III.3.6 and Example III.4.7])**.** *Let $P \in E(K)$. The translation $\tau_P\colon E \to E$, $Q \mapsto Q + P$, is an isomorphism defined over K, with inverse τ_{-P}.*

Problem 1.8 ($E(K)/E_0(K)$ is finite)**.**
Let $q\colon K^3 - \{0\} \to \mathbb{P}^2(K)$ be the quotient map. Give $K^3 - \{0\}$ the product topology and $\mathbb{P}^2(K)$ the quotient topology: $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U)$ is open. Let

$$S \coloneqq \{\mathbf{x} = (x_0, x_1, x_2) \in \mathcal{O}_K^3 : x_i \in \mathcal{O}_K^\times \text{ for some } i\}$$

be the set of primitive coordinate vectors.
(1) Show that S is compact and open in $K^3 - \{0\}$, and that $q(S) = \mathbb{P}^2(K)$. Deduce that $\mathbb{P}^2(K)$ is compact. Show that $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U) \cap S$ is open in S.[a]
(2) Show that $E(K) = V_+(F)(K)$ is closed in $\mathbb{P}^2(K)$, and deduce that it is compact.
(3) Let $g, h \in K[X, Y, Z]$ be homogeneous of the same degree. Show that $W \coloneqq \{Q \in \mathbb{P}^2(K) : h(Q) \neq 0\}$ is open, and that $W \to K$, $Q \mapsto g(Q)/h(Q)$, is continuous. Deduce from Propositions 1.6 and 1.7 that every translation $\tau_P\colon E(K) \to E(K)$ is a homeomorphism.
(4) Show that the reduction map $\mathbb{P}^2(K) \to \mathbb{P}^2(k)$, $P \mapsto \bar{P}$, is locally constant. Deduce that $E_0(K)$ is open in $E(K)$, and also closed.
(5) Prove that $E(K)/E_0(K)$ is finite.
(6) Let K' be a field complete with respect to a discrete valuation whose residue field k' is infinite, for instance algebraically closed. Show that its ring of integers is not compact. Which step of the argument above fails for an elliptic curve over K'?

[a] *Hint:* $q^{-1}(U)$ is stable under multiplication by $K^\times$.

2. Inertia acting on torsion

This section is [Sil09, Exercise 7.9], with the hypothesis $m \geqslant 3$ added to its first two parts (item (5) explains why) and a `Magma` check. We use the notation of [Sil09, §VII.4].

Definition 2.1 (Reduction types)**.** The curve E has good reduction if E_π is nonsingular, that is, if $v(\Delta) = 0$, and bad reduction otherwise. It has potential good reduction if it has good reduction over some finite extension of K.

Definition 2.2 (Inertia)**.** Let $\mathrm{Gal}_K := \mathrm{Gal}(\overline{K}/K)$, $K^{\mathrm{unr}} \subseteq \overline{K}$ the maximal unramified extension of K, and $I_v := \mathrm{Gal}(\overline{K}/K^{\mathrm{unr}})$ the inertia group. A set on which Gal_K acts is unramified if I_v acts trivially on it.

Definition 2.3 (Ramification of finite extensions)**.** For an extension L of K inside $\overline{K}$, let $\mathrm{Gal}_L := \mathrm{Gal}(\overline{K}/L)$. If L/K is finite and Galois, the inertia group of L/K is the image of I_v in $\mathrm{Gal}(L/K)$, isomorphic to $I_v/(I_v \cap \mathrm{Gal}_L)$. The extension L/K is unramified if its inertia group is trivial, and tamely ramified if its order is prime to p.

Definition 2.4 (Torsion fields and Tate modules [Sil09, §III.7])**.** For $m \geqslant 1$ with $p \nmid m$, $K(E[m])$ is the field generated over K by the coordinates of the points of $E[m](\overline{K})$, so $\mathrm{Gal}_{K(E[m])}$ is the kernel of the action of Gal_K on $E[m](\overline{K})$. For a prime $\ell \neq p$, the Tate module $T_\ell(E)$ is the inverse limit of the $E[\ell^n](\overline{K})$ along the surjections $[\ell]$, and $\rho_\ell \colon \mathrm{Gal}_K \to \mathrm{Aut}\, T_\ell(E)$ is the action of Gal_K on it.

In the following results, $\ell \neq p$ is a prime and $m \geqslant 1$ is an integer with $p \nmid m$.

Proposition 2.5 (Minimal equations [Sil09, Remark VII.1.1 and §III.1])**.** *A Weierstrass equation with coefficients in $\mathcal{O}_K$ and $v(\Delta) < 12$ is minimal. For $y^2 = x^3 + Ax + B$, $\Delta = -16(4A^3 + 27B^2)$.*

Corollary 2.6 (Torsion [Sil09, Corollary III.6.4(b)])**.** *The group $E[m](\overline{K})$ is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$.*

Proposition 2.7 (The Tate module [Sil09, Proposition III.7.1(a)])**.** *The Tate module $T_\ell(E)$ is isomorphic to $\mathbb{Z}_\ell^2$. Hence each projection $T_\ell(E) \to E[\ell^n](\overline{K})$ is a surjective homomorphism of Gal_K-modules, and its kernel is $\ell^n T_\ell(E)$, since both have index ℓ^{2n}.*

Proposition 2.8 (Good reduction [Sil09, Proposition VII.4.1])**.** *If E has good reduction, then $E[m](\overline{K})$ and $T_\ell(E)$ are unramified.*

Theorem 2.9 (Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1])**.** *If $T_\ell(E)$ is unramified, then E has good reduction.*

Corollary 2.10 (Potential good reduction [Sil09, Corollary VII.7.3])**.** *The curve E has potential good reduction if and only if $\rho_\ell(I_v)$ is finite.*

Proposition 2.11 (Inertia groups of finite extensions [Sil09, proof of Corollary VII.7.3])**.** *If $H \subseteq I_v$ is a closed normal subgroup of finite index, there is a finite extension K' of K inside $\overline{K}$ whose inertia group $\mathrm{Gal}(\overline{K}/K'K^{\mathrm{unr}}) = I_v \cap \mathrm{Gal}_{K'}$ is H.*

Problem 2.12 (The inertia group of $K(E[m])/K$)**.**
Suppose that E has potential good reduction, and let J be the subgroup of the elements of I_v that act trivially on $E[m](\overline{K})$ for every $m \geqslant 1$ with $p \nmid m$.
(1) Let ℓ be a prime and $r \geqslant 1$ with $\ell^r \geqslant 3$, and let $g \in \mathrm{GL}_2(\mathbb{Z}_\ell)$ be an element of finite order with $g \equiv I \pmod{\ell^r}$. Show that $g = I$.[a]
(2) Show that $J = I_v \cap \ker \rho_\ell$ for every prime $\ell \neq p$, and that J has finite index in I_v.
(3) Let $m \geqslant 3$ with $p \nmid m$. Show that $J = I_v \cap \mathrm{Gal}_{K(E[m])}$, and deduce that the inertia group of $K(E[m])/K$ is isomorphic to I_v/J, independently of m.
(4) Let $m \geqslant 3$ with $p \nmid m$. Show that $K(E[m])/K$ is unramified if and only if E has good reduction. Which direction holds for every $m \geqslant 1$ with $p \nmid m$, and for every elliptic curve over K?
(5) Let $p \geqslant 3$ and $E \colon y^2 = x^3 - \pi^2 x$.
(a) Show that E has bad reduction, potential good reduction, and $K(E[2]) = K$.
(b) Show that I_v/J has order 2, and that its nontrivial element acts on every $T_\ell(E)$ as -1.[b]
(c) So (3) and (4) fail for $m = 2$. Where did your proofs use $m \geqslant 3$?

(6) Suppose that $p \geqslant 5$. Show that $K(E[m])/K$ is tamely ramified for every $m \geqslant 1$ with $p \nmid m$.[c]
(7) `Magma` computes $\#(I_v/J)$ for elliptic curves over $\mathbb{Q}_7$:

```
p := 7;
Inertia := func< A, B |
  #InertiaGroup(GaloisRepresentation(EllipticCurve([A, B]), p)) >;
Inertia(-1, 0);                    // 1
Inertia(-p^2, 0);                  // 2
Inertia(0, p);                     // 6
```

Check the first two values against your answers, and the third against (6).

[a] *Hint:* reduce to g of prime order q, write $g = I + \ell^s h$ with $h \in M_2(\mathbb{Z}_\ell)$ not divisible by ℓ, and expand g^q.
[b] *Hint:* over $K(\sqrt{\pi})$, E is isomorphic to $y^2 = x^3 - x$.
[c] *Hint:* $\#\,\mathrm{GL}_2(\mathbb{F}_3) = 48$.

References

[Poo06] Bjorn Poonen, *Lectures on rational points on curves*, 2006. Lecture notes, March 5, 2006 version. ↑

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[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑